

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA0202

COURSE OUTCOME COVERED

CO3: Analyze shape representation methods and techniques such as contours, regions, deformable models, and multi-resolution analysis. (BL4)

Assessment Tool Weightage: 30%

QUESTIONS: 2 Total Marks:30 Annexure A

Case study

Code of Conduct:

I M Tilal Venkata Sai(192424415)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tilak Sai

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	

Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	
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Annexure A

Case study

Case Study 1: Defect Detection in Industrial Parts

A manufacturing company wants to automatically detect defects in gears and machine components using computer vision. The images of the components are binary and may contain small protrusions or missing parts.

Questions:

- How can region-based representation and medial axis computation help in identifying defective parts?
- Discuss how noise in image acquisition could affect shape analysis and propose preprocessing steps to improve accuracy.
- Compare the effectiveness of Fourier descriptors versus wavelet descriptors for classifying normal vs. defective components.

Case Study 2: Medical Image Segmentation

A hospital is developing a system to segment organs from CT scans. The organ boundaries are irregular and may vary slightly between patients. **Questions:**

- Explain how active contours (snakes) can be applied to segment the organ boundaries.
- Discuss the challenges of initializing contours and tuning parameters for accurate segmentation.
- How could level set methods improve segmentation for cases where organs split or merge in the images?

CASE STUDY 1: DEFECT DETECTION IN INDUSTRIAL PARTS

A. Region-Based Representation and Medial Axis

- Convert the binary component image into separate object regions using connected-component analysis.
- Measure region properties such as area, perimeter, bounding box and shape characteristics.
- Compute the medial axis or skeleton to obtain a thin central representation of the component.
- Compare the skeleton with a normal reference component. Missing parts can produce broken or shortened branches, while protrusions can create extra branches.
- Detect holes and abnormal regions by comparing region and skeleton properties with those of a defect-free component.

B. Effect of Noise and Preprocessing

- Noise can create false protrusions, holes, isolated components and unwanted skeleton branches.
- Apply thresholding to obtain a clean binary image before shape analysis.
- Use a median filter to reduce impulse noise while preserving important object boundaries.
- Apply morphological opening to remove small unwanted objects and closing to fill small gaps or holes.
- Use connected-component analysis to remove isolated noise regions before calculating the medial axis.
- These steps produce a cleaner shape and reduce false defect detection.

C. Fourier Descriptors vs. Wavelet Descriptors

- Fourier descriptors represent global contour shape using frequency components and are compact and computationally efficient.
- They are useful when defects cause significant changes in the overall boundary.
- Wavelet descriptors provide multi-resolution information and represent both coarse shape and localized changes.
- Wavelets are more suitable for detecting small or localized defects such as tiny protrusions or missing regions.
- Fourier descriptors are efficient for global comparison, while wavelet descriptors provide better local-detail analysis.

CASE STUDY 2: MEDICAL IMAGE SEGMENTATION

A. Application of Active Contours (Snakes)

- Active contours are curves that move toward an object's boundary by minimizing an energy function.
- Internal energy keeps the contour smooth, while external image forces attract it toward strong edges or intensity boundaries.
- Initialize a contour close to the organ boundary in the CT image.
- During iterations, the contour moves toward the actual organ boundary while maintaining a smooth shape.
- This method is useful for organs with irregular boundaries because the contour adapts to their shape.

B. Challenges in Initialization and Parameter Tuning

- The initial contour should be reasonably close to the organ. If it is too far away, the snake may converge to an incorrect structure.
- Weak or unclear organ boundaries can make contour convergence difficult.
- Smoothness parameters control how strongly the contour resists irregular deformation.
- Image-force parameters control how strongly the contour is attracted to visible boundaries.
- Too much smoothing can remove important details, while too little smoothing can produce irregular contours.
- Parameters should be selected and validated using representative CT images.

C. Level Set Methods for Splitting and Merging

- Level set methods represent an evolving boundary implicitly using a level-set function.
- Unlike a traditional snake, the contour can naturally split into multiple regions or merge into one region.
- Complex organ boundaries can therefore be handled without manually creating separate contours.
- Nearby regions can merge when the image forces favor a common boundary.
- Level set methods are more flexible than traditional active contours when the topology of the segmented object changes.